Key features of SCM

1. Inventory Management

Core Functionalities of an Inventory Management Module

- **Real-Time Stock Tracking & Visibility:** Provides a "single source of truth" for stock levels across multiple warehouses, retail stores, and items in transit. It tracks specific attributes like **SKUs**, **lot/batch numbers**, and **serial numbers** for full traceability.
- **Demand Forecasting & Planning:** Uses historical sales data, market trends, and predictive analytics (often AI-driven) to estimate future customer demand. This allows businesses to optimize stock levels and prepare for seasonal surges.
- **Automated Replenishment (Reordering):** Monitors stock against preset minimum thresholds. When levels fall below a specific point, the system automatically triggers alerts or generates **purchase orders** to prevent stockouts.
- **Inventory Optimization & Segmentation:** Employs techniques like **ABC Analysis** to categorize stock by value and turnover rate, allowing managers to prioritize high-value "A" items. It also calculates **Safety Stock** levels to buffer against supply chain disruptions.
- **Multi-Location Management:** Enables seamless tracking and **stock transfers** between different facilities or branches, ensuring inventory is moved to where it is most needed.
- **Lifecycle & Expiry Management:** Specifically tracks "shelf-life" for perishable goods. It uses **FIFO (First-In, First-Out)** or **FEFO (First-Expired, First-Out)** logic to minimize waste and ensure compliance.
- **Reporting & Financial Analytics:** Generates critical business insights such as **inventory turnover rates**, **carrying costs**, and **valuation reports** (e.g., LIFO, FIFO, or Average Cost).
- **Supplier & Vendor Management:** Centralizes supplier data, tracks lead times, and monitors performance to improve procurement reliability. Advanced systems may support **Vendor Managed Inventory (VMI)**, where suppliers directly monitor and replenish stock.

2. **<u>Warehouse Management</u>**
   real-time inventory tracking, barcode/RFID scanning, optimized receiving and put-away, and efficient picking/packing workflows.

   Core Functionalities of a WMS

- **Inbound Operations (Receiving & Putaway):**
  - **Receiving:** Validates incoming goods against purchase orders, logs item details (batch, serial numbers, expiration dates), and performs quality control checks.
  - **Putaway Management:** Uses algorithms to determine the best storage locations based on product size, weight, turnover frequency, and specialized requirements (e.g., temperature control).
  - **Cross-Docking:** Directly routes incoming goods from the receiving dock to shipping for immediate dispatch, bypassing long-term storage to reduce handling costs.
- **Inventory & Storage Management:**
  - **Real-Time Tracking:** Provides 100% visibility into stock levels and precise locations down to individual bins using barcode or RFID technology.
  - **Slotting & Layout Optimization:** Dynamically reallocates storage spots based on demand trends to reduce worker travel time and maximize space utilization.
  - **Cycle Counting:** Schedules regular, partial inventory counts to ensure system accuracy without shutting down warehouse operations.
  - **Replenishment:** Automatically triggers stock moves from bulk storage to active pick faces when levels drop below a threshold.
- **Outbound Operations (Fulfillment):**
  - **Order Picking:** Supports multiple strategies such as **Wave Picking** (scheduled groups), **Batch Picking** (picking for multiple orders at once), or **Zone Picking** (assigning workers to specific areas).
  - **Packing & Shipping:** Guides staff on correct box sizes, generates shipping labels, and integrates with carriers (e.g., FedEx, DHL) to automate documentation and tracking updates.
  - **Returns Management:** Streamlines reverse logistics by processing returned items, assessing their condition, and directing them for restocking, repair, or disposal.
- **Resource & Operational Management:**
  - **Labor Management:** Tracks employee performance against KPIs, forecasts staffing needs for peak periods, and automates shift scheduling.
  - **Yard & Dock Management:** Manages vehicle flow, schedules dock appointments, and tracks assets (like trailers) in the facility's exterior yard.

    - **Task Interleaving:** Orchestrates tasks so workers can perform activities like putaway on their return trip from a picking run, minimizing "deadheading" (empty travel).
- **Advanced & Technical Features:**
    - **Reporting & Analytics:** Provides dashboards for real-time KPIs such as order accuracy, inventory turnover, and fulfillment speed.
    - **Integration:** Connects seamlessly via APIs to other SCM components like **Transportation Management Systems (TMS)** and **Enterprise Resource Planning (ERP)** software.
    - **Automation Support:** Orchestrates advanced technologies like **Autonomous Mobile Robots (AMRs)**, automated conveyor systems, and voice-picking headsets.

3. Product Advancement and Returns Management

Product Advancement Functionalities

Product advancement within SCM focuses on the strategic evolution and development of a product from concept to commercialization. This module ensures that the supply chain is prepared for new launches and product upgrades.

- **Product Development & Commercialization:** Provides a structured framework for bringing new products to market jointly with suppliers and customers.
- **Product Lifecycle Management (PLM):** Manages all product-related data and processes from ideation and engineering through manufacturing and disposal.
- **Supplier Involvement in Design:** Facilitates early supplier involvement (ESI) to leverage external expertise, reducing concept-to-customer cycle times and improving design quality.
- **Roadmap & Milestone Tracking:** Enables product teams to craft roadmaps with date-based milestones and prioritize value-driven enhancements based on user feedback.
- **Traceability & Genealogy:** Monitors product history from raw materials to finished goods to ensure compliance and quality throughout its "advancement" stages.
- **Compliance & Regulatory Management:** Ensures that the latest regulatory practices are integrated into the product design and manufacturing phases.

Returns Management Functionalities

Returns management, or reverse logistics, handles the flow of goods back from the customer to the business, focusing on cost recovery and customer satisfaction.

- **Self-Service Customer Portals:** Branded online platforms where customers can independently initiate returns, choose shipping methods (drop-off or pickup), and print return labels.
- **Return Merchandise Authorization (RMA):** Automates the approval or rejection of return requests based on pre-defined company policies and rules.
- **Intelligent Disposition Logic:** Uses AI and data analytics to automatically decide if a returned item should be restocked, repaired, liquidated, or recycled to maximize value recovery.
- **Automated Tracking & Notifications:** Provides real-time tracking for customers and internal teams while sending automated status updates on refunds or exchanges.
- **Refund & Exchange Management:** Facilitates rapid processing of store credits, refunds, or product exchanges. Modern systems often incentivize exchanges over refunds to retain revenue.
- **Returns Analytics & Reporting:** Identifies SKU-level return trends and reasons for returns (e.g., sizing issues, defects) to inform product improvements and reduce future return rates.
- **Reverse Logistics Orchestration:** Manages the physical movement of goods back to the warehouse, including multi-carrier support and international customs documentation for cross-border returns

4. **Order fulfillment**

Core Functionalities of an Order Fulfillment Module

- **Order Capture & Centralization:** Automatically gathers orders from diverse sales channels—including e-commerce websites, mobile apps, EDI systems, and physical stores—into a single, unified view.
- **Inventory Promising (ATP/GTP):** Performs real-time **Available-to-Promise (ATP)** or **Global-to-Promise (GTP)** checks. It confirms product availability across all warehouses and provides customers with accurate delivery dates based on current stock and lead times.
- **Intelligent Order Routing & Allocation:** Dynamically determines the best fulfillment location (e.g., the closest warehouse with stock) to minimize shipping costs and transit time.
- **Order Validation & Fraud Detection:** Checks customer details, shipping addresses (often against postal databases), and payment authorizations to prevent errors or fraudulent transactions before processing begins.
- **Picking & Packing Orchestration:** Generates automated pick lists and packing slips. It may use AI to suggest optimal picking routes in the warehouse and determine the best packaging size to reduce "dimensional weight" shipping costs.
- **Shipping & Carrier Management:** Integrates with major carriers (e.g., FedEx, UPS, DHL) to compare real-time shipping rates, select the most cost-effective method, and automatically generate shipping labels and tracking numbers.
- **Real-Time Tracking & Communication:** Sends automated, branded updates to customers at every stage (order confirmed, shipped, out for delivery) and provides live tracking links to reduce "Where Is My Order" (WISMO) inquiries.
- **Returns & Reverse Logistics:** Simplifies the return process by providing self-service customer portals, generating return labels, and managing the inspection and restocking of returned goods.
- **Exception & Backorder Management:** Proactively identifies at-risk orders (e.g., delays due to supply shortages) and allows managers to re-prioritize high-value customers or handle partial shipments and backorders.

5. <u>Reporting and Analytics Dashboard</u>

Core Functionalities of the Dashboard Module

- **Real-Time Data Visualization:** Displays KPIs through interactive charts, graphs, heat maps, and gauges to provide an "at-a-glance" view of organizational health.
- **Drill-Down Capabilities:** Allows users to click on summary data points to explore underlying granular details, such as specific SKUs, individual warehouses, or carrier performance.
- **Automated Report Generation & Scheduling:** Reduces manual labor by automatically generating periodic (daily, weekly, monthly) reports and distributing them to relevant stakeholders.
- **Predictive & Prescriptive Analytics:** Uses machine learning to forecast demand, predict potential delivery delays, and suggest optimal courses of action (e.g., re-routing a shipment to avoid weather disruptions).
- **Exception Management & Alerts:** Triggers instant notifications for critical deviations, such as stock falling below safety levels, shipment delays, or supplier non-compliance.
- **Cross-Functional Data Integration:** Consolidates data from disparate sources—including ERP, WMS, TMS, and external IoT sensors—into a single source of truth.
- **Role-Based Customization:** Tailors views for different user levels:
    - **Strategic:** High-level goals (e.g., total revenue, market share) for executives.
    - **Tactical:** Departmental performance (e.g., procurement trends) for middle managers.
    - **Operational:** Real-time day-to-day metrics (e.g., current picking status) for floor supervisors.

Key SCM Metrics Tracked

Category	Key Performance Indicators (KPIs)
Inventory	Turnover ratio, days of supply on hand, carrying costs, and obsolete stock percentage.
Fulfillment	Perfect Order Rate, on-time delivery (OTD), order cycle time, and fill rate.
Procurement	Supplier lead time variability, supplier defect rate, and total spend analysis.
Logistics	Freight cost per unit, carrier on-time performance, and fuel cost fluctuations.

Sustainability	Carbon emissions per shipment and resource utilization efficiency.

6. **Transport and Logistics Management - Advanced Routing, Drop-Shipping, etc.**

In Supply Chain Management (SCM) software, the **Transport and Logistics Management** module—often referred to as a **Transportation Management System (TMS)**—acts as the primary platform for planning, executing, and optimizing the physical movement of goods.

Core Transport & Logistics Functionalities

- **Carrier Management & Selection:** Automates the process of comparing carrier rates, service levels (speed, reliability), and contract terms to select the most cost-effective shipping partner for each load.
- **Shipment Planning & Load Building:** Optimizes how goods are consolidated into shipments to maximize vehicle capacity, reduce "empty miles," and lower overall fuel consumption.
- **Real-Time Shipment Visibility:** Integrates with GPS and **IoT sensors** to provide live tracking of goods in transit, allowing managers and customers to monitor status and receive updated ETAs.
- **Freight Audit & Payment:** Automatically verifies carrier invoices against negotiated contracts to ensure billing accuracy and streamline the settlement process.
- **Exception Management:** Detects deviations from planned routes or schedules (due to weather, traffic, or delays) and triggers automated alerts or contingency responses.

Advanced Routing Functionalities

Advanced routing goes beyond simple point-to-point navigation by using AI and machine learning to manage complex delivery variables.

- **Dynamic Route Optimization:** Recalculates delivery paths in real-time based on live traffic, road closures, and new order arrivals to minimize travel time and fuel costs.
- **Multi-Stop & Multi-Modal Routing:** Plans efficient paths involving multiple pickups and deliveries, often coordinating across different modes of transport (air, sea, rail, and truck).
- **Constraint-Aware Planning:** Factors in specific limitations such as vehicle weight/height restrictions, driver shift hours, and strict customer delivery windows.
- **Territory & Zone Management:** Automatically assigns orders to specific geographical "service zones" to improve driver familiarity and route density.

Drop-Shipping Functionalities

Within SCM logistics, drop-shipping shifts the fulfillment burden to a third party, requiring specific digital coordination features.

- **Order Orchestration:** Automatically identifies "third-party" items in a sales order and triggers a **Purchase Requisition (PR)** or **Purchase Order (PO)** to the supplier instead of a warehouse pick-list.

- **Supplier Portal Integration:** Provides a platform for suppliers to receive order details, print branded packing slips, and upload **Advance Shipment Notifications (ASN)**.
- **Automated Tracking Sync:** Captures tracking numbers from the supplier and automatically updates the customer, maintaining the "face" of the original seller.
- **Reverse Logistics for Dropshipping:** Manages the specialized return flow where customers ship items back to the original supplier, while the software handles the financial reconciliation (refunds/credits) between all three parties

7. **Real-Time Delivery Monitoring**

In Supply Chain Management (SCM) software, a real-time delivery monitoring module provides a "digital nervous system" for logistics, using technologies like GPS, IoT, and AI to track goods continuously from the warehouse to the customer's doorstep.

Core Tracking & Monitoring Functionalities

- **Live Asset & Location Tracking:** Uses GPS and satellite-based systems to provide precise, up-to-the-minute coordinates of vehicles, containers, or individual parcels on interactive digital maps.
- **Environmental Condition Monitoring:** Leverages IoT sensors to track critical factors such as **temperature** (for cold chains), **humidity**, **shock/vibration**, **light exposure**, and **pressure**.
- **Predictive Estimated Time of Arrival (ETA):** Employs machine learning to calculate highly accurate ETAs by factoring in real-time traffic, weather patterns, and historical delivery performance.
- **Geofencing & Automated Alerts:** Sets virtual boundaries that trigger instant notifications when a shipment enters or exits specific zones (e.g., ports or warehouses) or deviates from a planned route.

Operational & Compliance Functionalities

- **Proactive Exception Management:** Automatically detects and alerts managers to anomalies such as unplanned stops, unauthorized door access, or unexpected delays, allowing for immediate intervention.
- **Electronic Proof of Delivery (ePOD):** Captures digital signatures, timestamps, and photographs at the moment of delivery, which are immediately synced back to the central system for billing and audit purposes.
- **Fleet & Driver Performance Tracking:** Monitors driver behavior (e.g., harsh braking, speeding, or excessive idling) and vehicle health to improve safety and optimize fuel efficiency.
- **Historical Data Playback:** Provides a digital trail of past shipment routes and conditions, which is essential for resolving customer disputes, insurance claims, and regulatory audits.

Customer Engagement Functionalities

- **Self-Service Tracking Portals:** Provides branded web or mobile interfaces where customers can view their order's live status without contacting support.
- **Automated Status Notifications:** Sends real-time updates via SMS or email at key milestones, such as "out for delivery" or "successfully delivered".

8. **<u>Demand Forecasting</u>**
   The demand forecasting module in Supply Chain Management (SCM) software serves as the predictive engine for the entire organization, using historical data and market signals to estimate future customer needs.

   Core Module Functionalities

- **Statistical Baseline Generation:** Automatically creates initial forecasts using historical transaction data and standard time-series methods like **moving averages** or **exponential smoothing**.
- **AI & Machine Learning Engines:** Employs advanced algorithms to identify complex, non-linear patterns—such as "hidden" demand signals—that traditional statistics might miss.
- **Demand Sensing:** Captures real-time signals from point-of-sale (POS) data, social media trends, and weather patterns to refine short-term, near-term forecasts.
- **Collaborative Planning (CPFR):** Provides a shared platform for sales, marketing, and finance teams to contribute "soft" intelligence (e.g., upcoming promotions or market insights) into a single unified forecast.
- **Scenario & "What-If" Analysis:** Allows planners to simulate various hypothetical situations, such as price changes, new competitor entries, or supply disruptions, to assess their impact on demand.
- **Multivariate Regression:** Analyzes the relationship between demand and multiple external variables simultaneously, such as interest rates, GDP, or competitor pricing.

Operational Control Features

- **Data Cleansing & Outlier Removal:** Automatically identifies and "smooths out" one-time anomalies—like a celebrity endorsement spike or a natural disaster dip—to prevent them from skewing future projections.
- **Multi-Level/Dimensions Hierarchies:** Generates forecasts at various granularities, including by **SKU**, **geographic region**, **customer segment**, or **fulfillment channel**.
- **Forecast Accuracy Monitoring:** Tracks key performance metrics such as **Mean Absolute Percentage Error (MAPE)** and **forecast bias** to continuously refine and improve the model.
- **Authorization & Lifecycle Management:** Includes workflows for reviewing and manually adjusting baseline forecasts before they are officially "authorized" for use in production and procurement planning.
- **Promotion & Event Management:** Specifically models the "lift" expected from planned marketing campaigns or holiday events, automatically adjusting recommended order quantities.

9. Multi-level Pricing & Discount Management

In SCM software, the **Multi-level Pricing & Discount Management** module is the financial engine that ensures price consistency across complex global supply chains. It moves beyond "one price fits all" to dynamic, rule-based pricing tailored to specific customer segments, volumes, and contract terms.

Core Pricing Functionalities

- **Hierarchical Price Lists:** Manage multiple price tiers based on the buyer's profile (e.g., Distributor vs. Wholesaler vs. End-Retailer). This ensures that each layer of the supply chain sees only the pricing relevant to their tier.
- **Contract-Specific Pricing:** Stores and applies negotiated rates for specific long-term partners, overriding general market prices once a customer is identified in the system.
- **Multi-Currency & Regional Management:** Automatically converts prices based on real-time exchange rates and applies localized taxes or regional surcharges (e.g., Oracle Price Management functionality).
- **Dynamic/Real-Time Pricing:** Adjusts prices instantly based on external market triggers, such as fluctuating raw material costs (commodity prices) or competitor price changes.
- **Landed Cost Calculation:** Integrates shipping, duties, and insurance into the final price to ensure profit margins are protected during cross-border transactions.

Discount & Incentive Management

- **Volume & Tiered Discounts:** Automatically applies lower per-unit prices as order quantities increase (e.g., $10/unit for 1-100 units, $8/unit for 101+).
- **Promotional & Seasonal Rule Engines:** Sets "start" and "end" dates for marketing-led discounts, ensuring they apply only during specific campaigns or holiday windows.
- **Bundled Pricing:** Manages discounts for specific product combinations (e.g., "Buy a Printer and 3 Ink Cartridges, get 15% off").
- **Rebate & Loyalty Programs:** Tracks cumulative purchases over a quarter or year and calculates "back-end" rebates to be paid out or credited to the customer once targets are met.
- **Coupon & Referral Code Tracking:** Validates and tracks the performance of specific discount codes used across various sales channels.

Governance & Strategy

- **Approval Workflows:** Triggers a management review if a sales representative attempts to apply a discount that exceeds a pre-defined threshold.
- **Margin Protection Alerts:** Provides real-time visibility into the "floor price," preventing the system from accepting orders where the discount would result in a negative profit margin.
- **Profitability Analytics:** Generates reports showing which pricing strategies (e.g., "Buy 1 Get 1") are driving the most revenue versus which are eroding margins.

10. **Multi-tier Commission tracking**

In SCM software, the **Multi-tier Commission Tracking** module acts as the financial engine for complex sales networks, managing payouts for internal sales teams, external agents, distributors, and referral partners. It ensures that every stakeholder in a multi-layered supply chain is compensated accurately based on their contribution to a sale.

Core Commission Configuration

- **Hierarchical Structure Mapping:** Defines "parent-child" relationships in the sales org (e.g., Regional Manager > Territory Manager > Sales Rep) to ensure commissions flow upward correctly (overrides).
- **Variable Rate Logic:** Sets different commission percentages based on product category, profit margins, or customer type (e.g., 5% for new customers, 2% for renewals).
- **Threshold-Based Triggers:** Automates "accelerators" where commission rates increase once a rep hits 100% of their quota or a specific revenue milestone.
- **Split Commission Management:** Facilitates the sharing of a single commission between multiple parties, such as when a lead is generated by one person but closed by another.

Payout & Execution Functionalities

- **Automated Calculation Engines:** Syncs with the Order Management Module to calculate payouts in real-time as orders are invoiced or payments are received.
- **Clawback Automation:** Automatically reverses or "claws back" commissions if an order is returned, cancelled, or if the customer fails to pay (bad debt).
- **Draw-Against-Commission:** Manages "draws" (advance payments) for new hires or seasonal reps, automatically deducting the draw amount from future earned commissions.
- **Multi-Currency Support:** Calculates and converts commission payouts for global teams based on the exchange rate at the time of the transaction.

Transparency & Reporting

- **Rep Self-Service Dashboards:** Provides individual sales agents with a real-time view of their "estimated" vs. "earned" commissions and progress toward their next tier.
- **Audit Trail & Compliance:** Maintains a detailed log of every calculation and adjustment, which is critical for financial audits and resolving payout disputes.
- **Dispute Management Workflow:** Provides a formal channel within the software for reps to flag and resolve discrepancies in their payout statements.
- **Statement Generation:** Automatically produces detailed monthly or quarterly commission statements that can be exported for payroll or tax purposes.

Strategic Integration

- **KPI & Performance Analytics:** Generates reports on which commission structures are driving the most high-margin sales versus high-volume/low-margin sales.
- **Payroll Integration:** Seamlessly pushes finalized commission data to ERP or payroll systems (like SAP SuccessFactors or Workday) to ensure timely payment.

11. Barcode Reader and RFID Tracker

In Supply Chain Management (SCM) software, the **Barcode Reader and RFID Tracker** module serves as the primary gateway for digitalizing physical inventory movements. While barcodes are a cost-effective standard for individual item identification, RFID (Radio Frequency Identification) provides advanced bulk-tracking and non-line-of-sight capabilities.

Core Functionalities of the Module

- **Automated Data Capture & Identification:** Converts physical symbols (1D/2D barcodes) or radio signals (RFID) into digital data, identifying products by SKU, serial number, batch, or lot.
- **Inbound & Outbound Processing:**
    - **Receiving & Verification:** Instantly matches incoming goods against purchase orders to verify quantities and flag discrepancies.
    - **Picking & Packing:** Validates that the correct items are selected for orders, significantly reducing mis-picks.
    - **Shipping Documentation:** Automatically generates shipping labels, manifests, and tracking numbers upon scanning the final package.
- **Inventory & Asset Tracking:**
    - **Real-Time Stock Updates:** Synchronizes physical moves (like bin-to-bin transfers) with the central database immediately.
    - **Rapid Cycle Counting:** RFID allows for "bulk scanning," enabling staff to count hundreds of items in seconds by simply walking through an aisle.
    - **Asset Lifecycle Management:** Tracks the location and maintenance history of fixed assets like forklifts, laptops, or reusable shipping containers.
- **Operational Security & Compliance:**
    - **Loss Prevention:** RFID "geofencing" can trigger alarms if high-value items move toward unauthorized exits.
    - **Traceability & Recall Management:** Maintains a "digital fingerprint" for every item, essential for tracking expiration dates or executing targeted product recalls.
- **Advanced Environmental Monitoring (RFID only):** Specialized RFID tags can monitor and log temperature or humidity, which is critical for cold chain compliance

12. Double-Entry Stock Keeping

In an SCM software, the **Double-Entry Stock Keeping** module treats every inventory transaction as a movement between two locations, similar to double-entry accounting. Instead of just adding or subtracting from a total, the system records an "output" from one location and an "input" to another, ensuring the entire supply chain remains balanced.

Core Functionalities of Double-Entry Stock Keeping

- **Location-Based Stock Moves:** Every operation—including receiving, internal transfers, or sales—is recorded as a transfer between physical or **virtual locations**. For example, receiving 20 units from a supplier logs them leaving a "Supplier" virtual location and entering your "Warehouse".
- **Virtual Location Management:** Uses virtual locations to represent entities outside the physical warehouse, such as:
    - **Partner Locations:** Representing suppliers and customers.
    - **Production Locations:** Tracking raw materials "consumed" and finished goods "produced".
    - **Inventory Adjustment Locations:** Used for handling stock losses, theft, or scrapped items by "moving" them into a specific loss account.
- **Native Accuracy & Error Detection:** Since every debit from one location must have a corresponding credit to another, the system can instantly identify **mismatched entries**, making it easier to detect missing stock or illegal traffic in real time.
- **Advanced Traceability:** Provides a comprehensive "genealogy" of every product, tracking its exact path through various nodes using serial numbers or lot tracking.
- **Real-Time Inventory Valuation:** Automatically calculates the value of stock at any specific location by multiplying quantities by their cost, providing instant insights into **carrying costs** and **value creation**.
- **Automated Procurement Routes:** Uses "push" and "pull" rules to automate stock replenishment and define complex paths a product must follow across multiple warehouses or steps.
- **Scrap & Loss Management:** Allows products to be "scrapped" directly from receipts, delivery orders, or internal transfers by moving them to a dedicated scrap location.
- **Financial Integration:** Seamlessly syncs operational stock moves with the **General Ledger**, turning physical movements into automated financial journal entries for accounting.

13. Delivery Orders

In supply chain management (SCM) software, the **delivery orders (DO) module** (sometimes referred to as the delivery challan or cargo release module) manages the final authorization and physical execution of moving goods to the customer. It bridges the gap between sales confirmation and physical delivery.

Core Functionalities of the Delivery Orders Module

- **Order Authorization & Cargo Release:** Acts as the official "green light" to a warehouse or terminal operator to release specific goods to a verified consignee.
- **Documentation Generation:** Automatically creates standardized delivery orders/challans that include:
    - **Unique Tracking Identifiers:** Delivery note or challan numbers.
    - **Detailed Item Lists:** Product names, quantities, HSN codes, and unit weights.
    - **Logistics Data:** Carrier details, vehicle numbers, E-Way bill numbers, and dispatch locations.
- **Automated Scheduling & Dispatch:** Assigns delivery tasks to the most suitable drivers or vehicles based on predefined rules like volume, geography, and shift hours.
- **Real-Time Tracking & Visibility:** Provides live GPS tracking of the delivery vehicle's location and progress, offering customers accurate ETAs and proactive delay notifications.
- **Proof of Delivery (POD) Capture:** Digitally records the successful handover of goods through mobile signatures, photos of the delivered items, and GPS-stamped coordinates.
- **Inventory Synchronization:** Automatically decreases inventory balances once a delivery note is recorded, ensuring stock levels reflect actual physical departures.
- **Advanced Route Optimization:** Uses algorithms to calculate the most efficient delivery paths, factoring in real-world constraints such as traffic, road closures, and delivery time windows.
- **Status Management & Billing Integration:** Tracks the lifecycle of the delivery (e.g., Draft → Authorized → Shipped → Delivered). Once marked "Delivered," it triggers the generation of the final sales invoice or billing document.

14. Packing, Repairs, Receipts Management

In Supply Chain Management (SCM) software, these modules work together to ensure products are correctly prepared for transit, maintained throughout their lifecycle, and accurately documented for financial and inventory purposes.

1. Packing Module Functionalities

This module automates the process of preparing goods for shipment to ensure efficiency and product safety.

- **Cartonization & Configuration:** Determines the optimal box or pallet configuration based on product dimensions and weight to minimize shipping costs and maximize space.
- **Labeling & Traceability:** Generates compliant shipping labels, barcodes, and RFID tags for real-time tracking and identification throughout the supply chain.
- **Protective Specification:** Defines specific packaging requirements for fragile, temperature-sensitive, or hazardous items, such as moisture resistance or physical padding.
- **Unitization:** Groups individual products into larger secondary or tertiary units (e.g., cases or pallets) for easier handling by both manual labor and automated systems.
- **Sustainability Tracking:** Manages the use of eco-friendly materials and identifies opportunities to reduce packaging waste.

2. Repairs Management Module Functionalities

This module streamlines the entire maintenance and repair lifecycle, reducing equipment downtime and service costs.

- **Work Order Management:** Creates and tracks repair requests from initial submission to final completion.
- **Technician Assignment:** Automatically assigns tasks to the best-suited technician based on availability, skill set, and location.
- **Spare Parts Tracking:** Deeply integrates with inventory to monitor part consumption and trigger automated reordering when levels are low.
- **Warranty & Claim Management:** Automatically verifies warranty status for parts and labor before initiating repairs to ensure accurate billing.
- **Predictive Maintenance:** Uses AI and sensor data to forecast potential failures and schedule maintenance before breakdowns occur.
- **Cost Estimation:** Calculates the total cost of a repair, including labor hours, parts used, and travel expenses.

3. Receipt Management Module Functionalities

This module digitizes and organizes transaction documentation for financial accuracy and audit readiness.

- **Digital Capture (OCR):** Uses Optical Character Recognition to extract data (vendor, amount, date) from photos or scans of physical receipts.

- **Automated Categorization:** Tags receipts by expense type, project, or department to simplify budgeting and reporting.
- **3-Way Matching:** Automatically compares receipt data against purchase orders and invoices to verify that the goods were actually received as ordered.
- **Accounting Integration:** Syncs receipt data directly with financial software (e.g., QuickBooks, SAP) to eliminate manual data entry.
- **Cloud Storage & Retrieval:** Provides a secure, searchable archive of all digital receipts for tax compliance and audit preparation.
- **Reimbursement Processing:** Streamlines employee expense reports by digitizing receipts on-the-go via mobile apps.

15. <u>Database design</u>

Data Storage & Integration

- Maintains **master data** for suppliers, customers, products, warehouses, and logistics partners.
- Integrates information across procurement, production, inventory, and distribution.
- Ensures **single source of truth** for all supply chain activities.

Inventory Management

- Tracks **stock levels, reorder points, and safety stock**.
- Supports **real-time updates** on product availability.
- Helps avoid **stockouts or overstocking** by linking with demand forecasting.

Order & Shipment Tracking

- Stores details of **purchase orders, sales orders, and shipping documents**.
- Enables **end-to-end visibility** of shipments from supplier to customer.
- Provides alerts for **delays, discrepancies, or exceptions**.

Supplier & Customer Relationship Management

- Maintains supplier performance records (lead times, quality, reliability).
- Stores customer order history and preferences.
- Facilitates **better negotiation and collaboration** with partners.

Analytics & Reporting

- Generates **KPIs** such as order fulfillment rate, inventory turnover, and supplier reliability.
- Provides **dashboards and reports** for decision-making.
- Supports **predictive analytics** for demand forecasting and risk management.

6. **Security & Compliance**

- Ensures **data integrity and confidentiality**.
- Supports compliance with **industry standards and regulations** (e.g., ISO, customs documentation).
- Provides **audit trails** for accountability.

16. <u>UI for the whole application</u>

Dashboard & Visualization

- Provides **real-time dashboards** for inventory, orders, shipments, and supplier performance.
- Uses **charts, graphs, and KPIs** to simplify complex data.
- Enables quick decision-making through **visual alerts and indicators**.

Navigation & Workflow Management

- Offers **intuitive menus and navigation paths** for procurement, logistics, and distribution modules.
- Guides users through **step-by-step workflows** (e.g., creating purchase orders, tracking shipments).
- Reduces training time with **user-friendly layouts**.

Data Entry & Forms

- Simplifies **transaction entry** (orders, invoices, receipts).
- Provides **validation checks** to reduce errors during input.
- Supports **bulk uploads and automated form filling**.

Role-Based Access & Customization

- Displays **personalized views** based on user roles (procurement officer, warehouse manager, logistics coordinator).
- Allows customization of **dashboards, reports, and shortcuts**.
- Ensures **security by restricting sensitive data** to authorized users.

Notifications & Alerts

- Provides **pop-up alerts, emails, or system messages** for critical events (stockouts, shipment delays, order failures).
- Enables proactive management by highlighting **exceptions**.

Integration with Other Modules

- Connects seamlessly with **database, error handling, and analytics modules**.
- Presents consolidated information from multiple sources in a **single interface**.

17. Excepption handling

Error Detection & Logging

- Identifies failures in **transactions, workflows, or integrations**.
- Logs errors with detailed metadata (time, process step, affected entity).
- Supports **real-time monitoring dashboards** for visibility.

Rollback & Data Consistency

- If an action fails, the system **resets or rolls back** previously executed steps to maintain consistency.
- Prevents partial updates that could corrupt supply chain records.
- Ensures that only fully successful actions are posted to the database.

Error Queues & Workflow Continuity

- Failed actions are placed in an **error queue** for review and reprocessing.
- Allows subsequent actions to continue while isolating the failed step.
- Minimizes disruption to overall supply chain workflows.

Notification & Escalation

- Automatically informs **responsible users or teams** when errors occur.
- Supports configurable **alerts, emails, or system messages**.
- Escalates unresolved issues to higher authority levels.

Diagnostics & Root Cause Analysis

- Provides tools to analyze **system errors, orchestration failures, and integration mismatches**.
- Helps identify whether the issue is due to **data quality, system configuration, or external partner systems**

Reprocessing & Recovery

- Allows administrators to **reprocess failed transactions** after correction.
- Supports **manual overrides** or automated retries.
- Ensures smooth recovery without duplicating or losing data.